CSCI 5526

Causal Inference in Healthcare

[Slides based on MIT Machine Learning for Healthcare Lecture 14 by David Sontag]

Announcements

- MP2 due on Friday!
- Self-sign up for final project groups on Canvas!
  - Due on 10/16/26!

Student Questions

Is it even worth always finding the most complicated way to map the (EHR) data? Are simpler/cruder methods better?

Short answer is No!

Simpler methods are preferred because they can achieve 80-90% of the predictive power with high interpretability and low overhead.

Complex mapping is potentially useful for:

- Rare Disease Research
- Long-horizon temporal deep learning
- Natural Language Processing of unstructured clinical notes

Is risk stratification growing due to the growth in deep learning? I'd imagine with the data complexity and size it would help a lot, but how does the logistic regression perform so well at the same time?

Yes, especially for applications based on unstructured data!

For many EHR-based ML tasks, logistic regression would work well for the above reasons.

Student Questions

In the second diabetes diagnosis scenario, how do we account for time-varying factors, unobserved/latent variables that may arise during the gap between feature construction and outcome determination? Are we assuming that the features constructed at baseline adequately capture the patient's future trajectory?

Yes, correct. Feature construction strategy is crucial and will determine the maximum potential predictive capacity!

How can continuous physiological data, such as heart rate, be effectively represented and encoded within an Electronic Health Record (EHR) system?

The EHR system typically does not encode continuous data streams, it only includes compressed summary metrics.

Continuous data are typically stored in timeseries databases.

Student Questions

If a risk stratification model is good at identifying high-risk patients, how can we know that using that model to intervene actually improves their health outcomes?

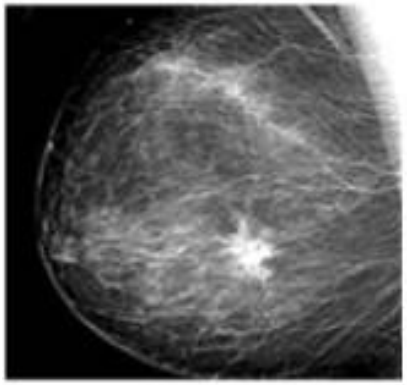
By conducting rigorous prospective clinical evaluations, such as randomized controlled trials (RCTs).

Carefully evaluate Association vs. Modifiability.

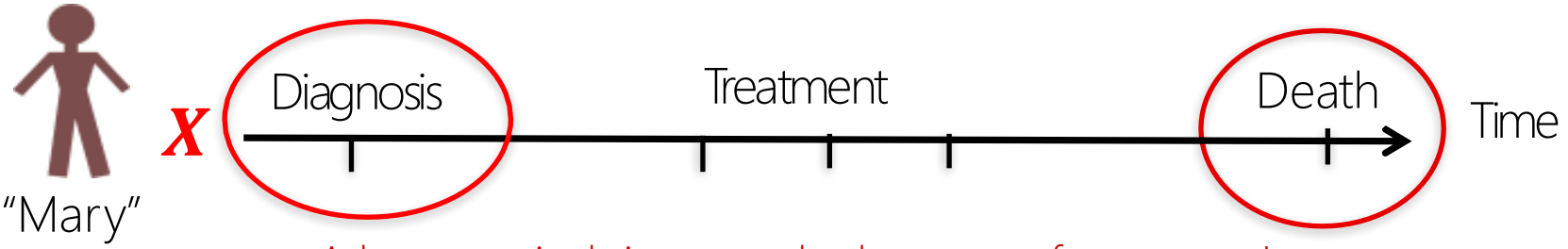
Outline

- Causal inference: examples in healthcare
- Formulating the problem
- The potential outcomes framework
- An illustrative example
- Assumptions
- Computational solutions

Survival in Breast Cancer



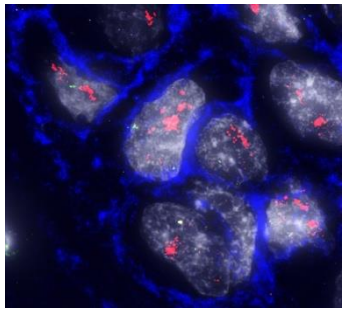
- Predictive models widely used to stage patients. Should we initiate treatment? How aggressive?
- What could go wrong if we trained to predict survival, and then used to guide patient care?



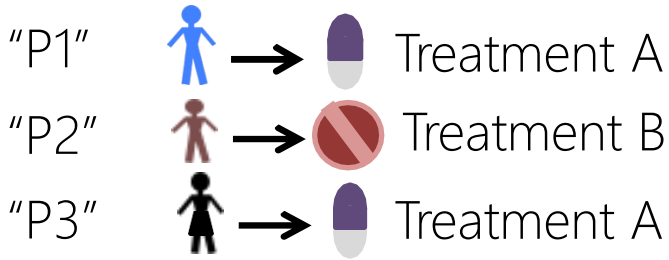
A long survival time may be because of treatment!

What treatment should we give this patient?

- People respond differently to treatments.
- Goal: use data from other patients and their journeys to guide future treatment decisions.
- What could go wrong if we trained to predict (past) treatment decisions?



Expansion pathology
(image from Andy Beck)



Best this can do is match current medical practice!

Does smoking cause lung cancer?

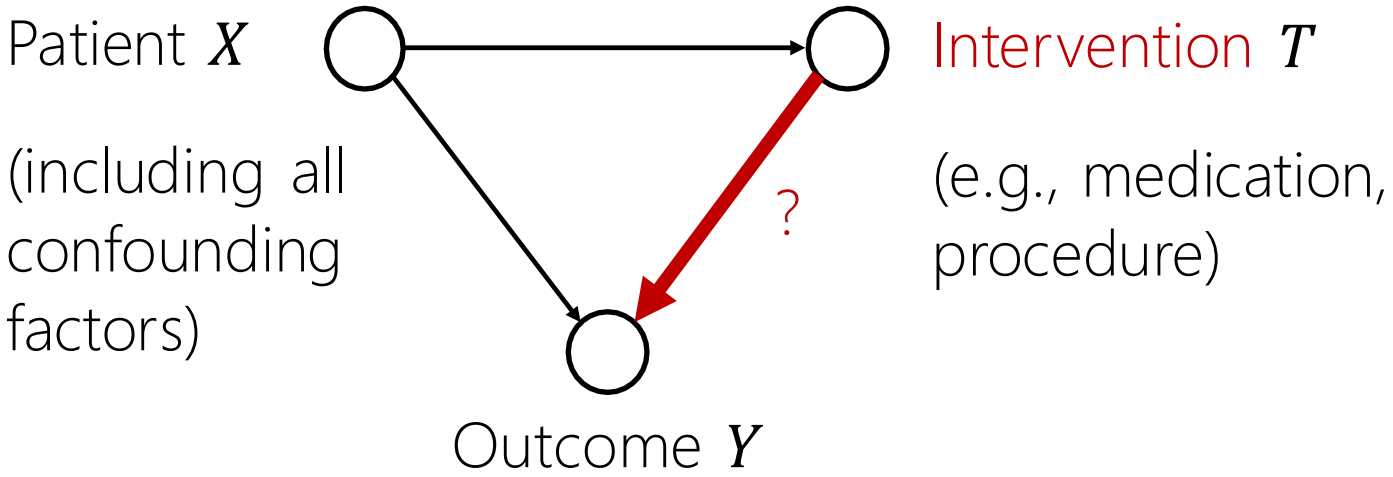


- Doing a randomized control trial is unethical.
- Could we simply answer this question by comparing P(lung cancer | smoker) vs P(lung cancer | nonsmoker)?
- No! Answering such questions from observational data is difficult because of confounding.

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Formulating the Causal Question



High dimensional

Observational data

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Potential Outcomes Framework (Rubin-Neyman Causal Model)

- Each unit (individual) x_i has two potential outcomes:
  - $Y_0(x_i)$ is the potential outcome had the unit not been treated: "control outcome"
  - $Y_1(x_i)$ is the potential outcome had the unit been treated: "treated outcome"

- Conditional average treatment effect for unit i:

$$CATE(x_i) = \mathbb{E}[(Y_1 - Y_0)|x_i] = \mathbb{E}[Y_1|x_i] - \mathbb{E}[Y_0|x_i]$$

- Average Treatment Effect:

$$ATE = \mathbb{E}[Y_1 - Y_0]$$

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$$CATE(x_i) = \mathbb{E}_{Y_1 \sim p(Y_1|x_i)}[Y_1|x_i] - \mathbb{E}_{Y_0 \sim p(Y_0|x_i)}[Y_0|x_i]$$

- Average Treatment Effect:

$$ATE = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}_{x \sim p(x)}[\mathbb{E}[(Y_1 - Y_0)|x]] = \mathbb{E}_{x \sim p(x)}[CATE(x)]$$

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- Observed factual outcome:

$$y_i = t_i Y_1(x_i) + (1 - t_i) Y_0(x_i)$$

t_i - treatment

- Unobserved counterfactual outcome:

$$y_i^{CF} = (1 - t_i) Y_1(x_i) + t_i Y_0(x_i)$$

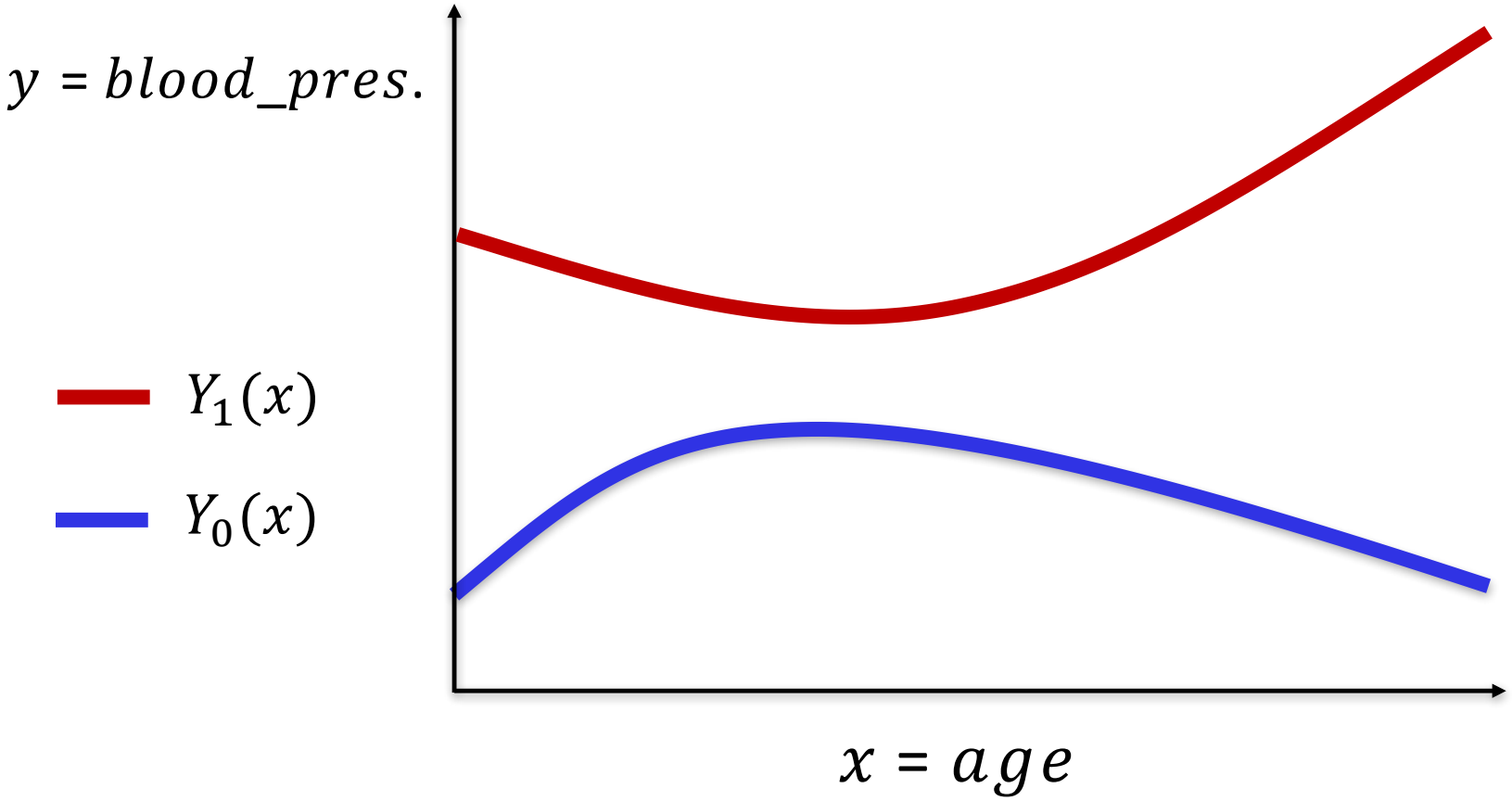
"The fundamental problem of causal inference"

We only ever observe one of the two outcomes

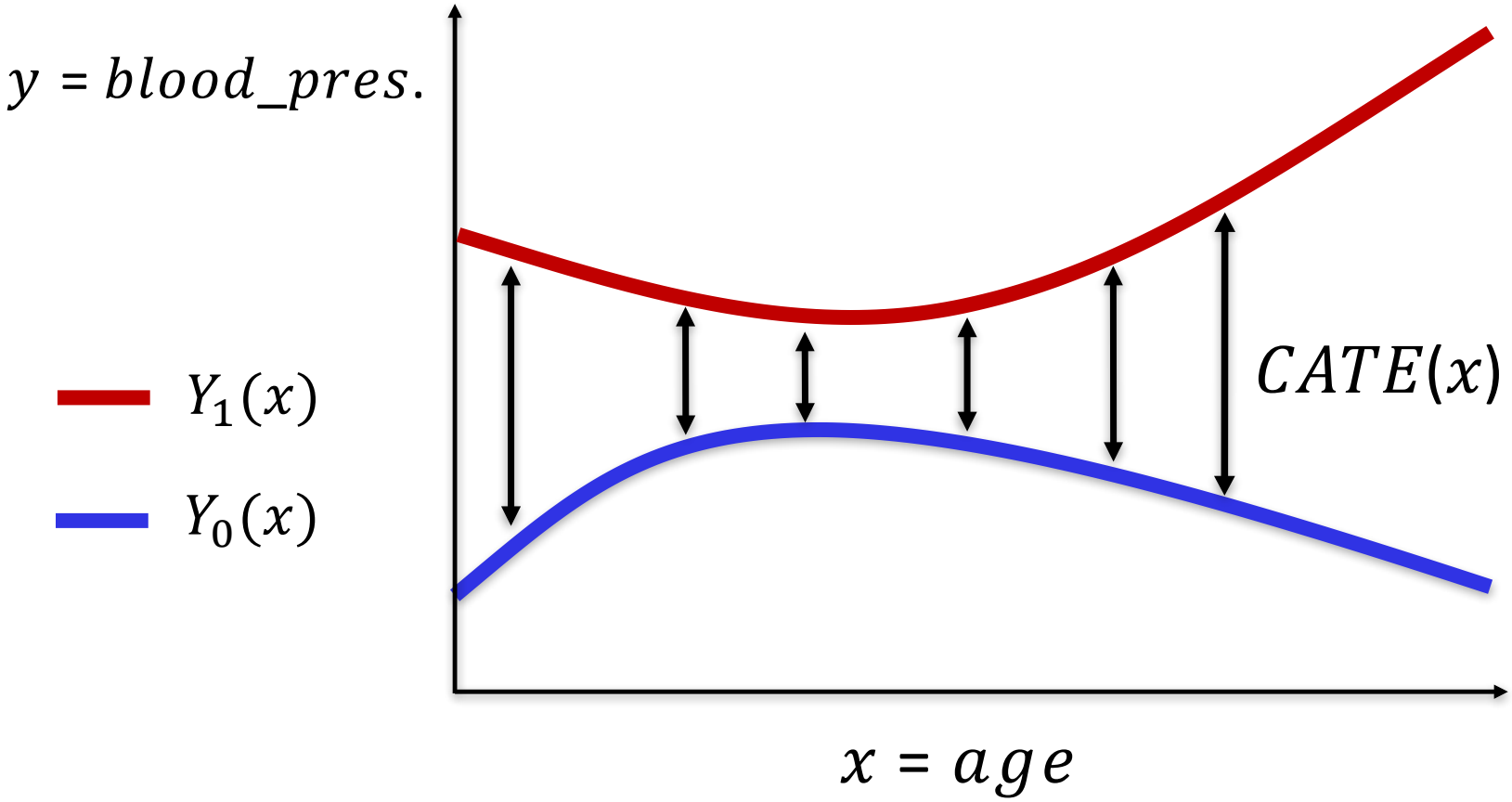
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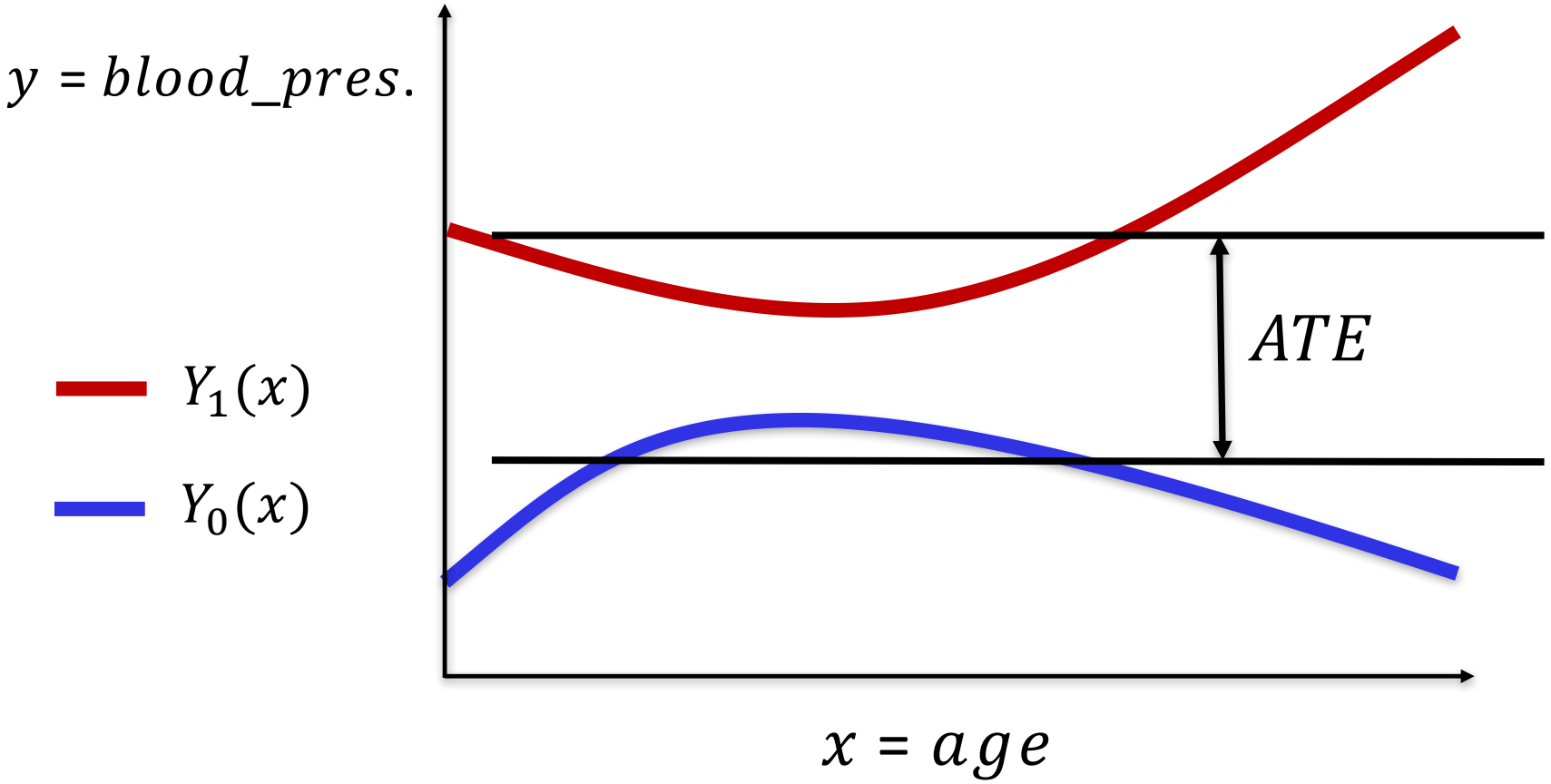
Example – Blood pressure and age



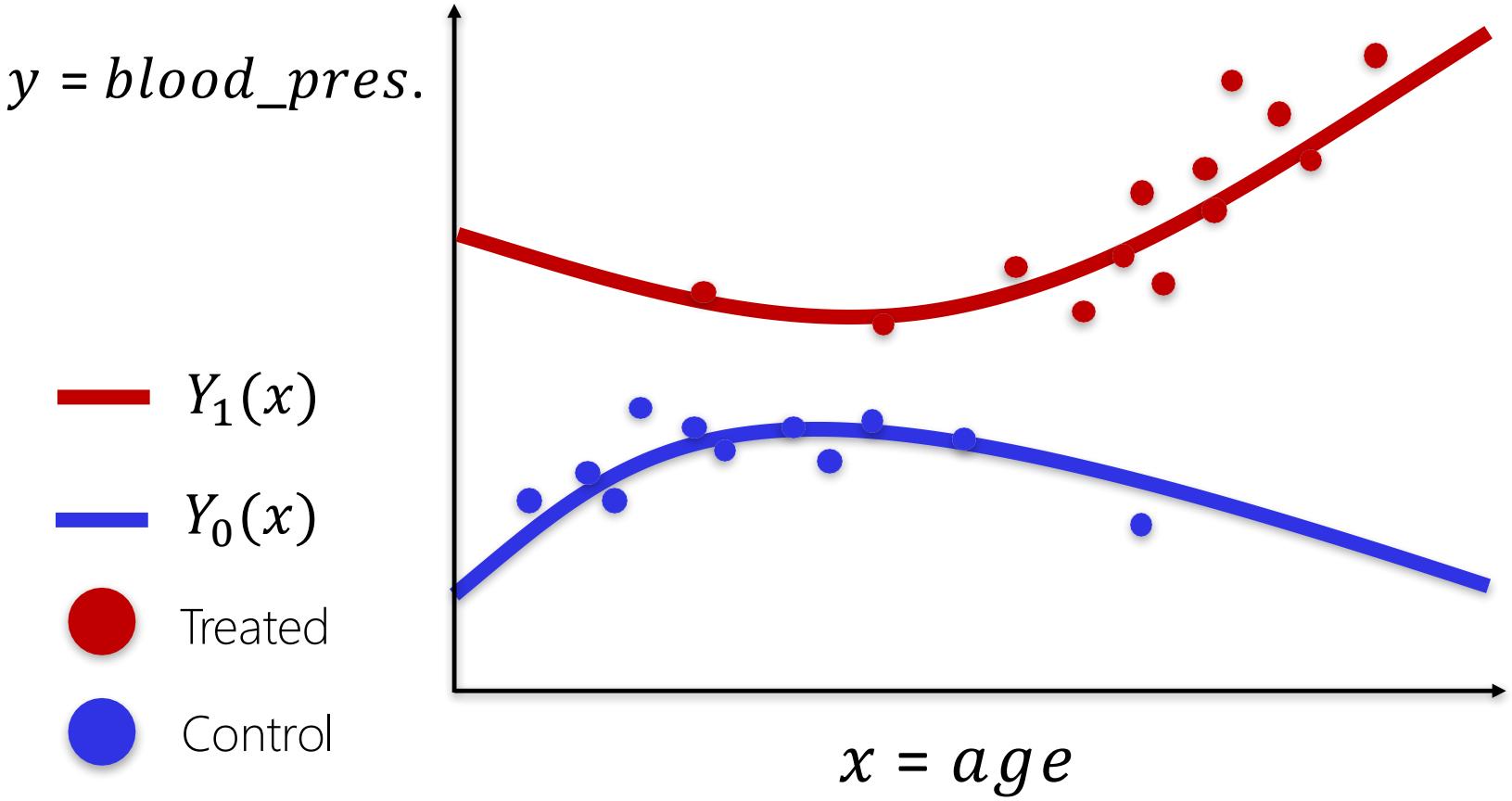
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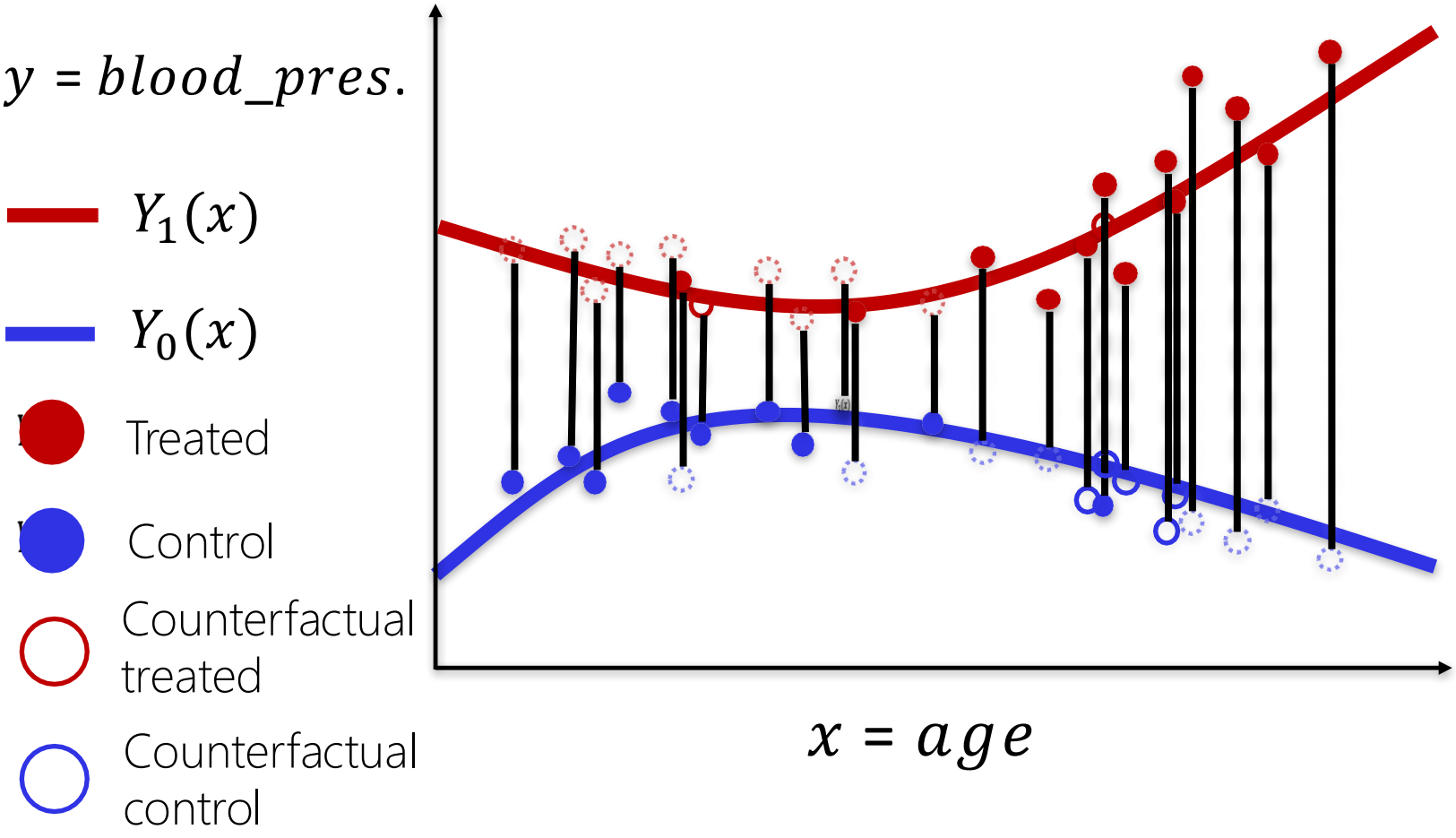


Example – Blood pressure and age



Example – Blood pressure and age

$y = blood_pres.$



(age, gender, exercise, treatment)			Observed sugar levels
(45, F, 0, A)			6
(45, F, 1, B)			6.5
(55, M, 0, A)			7
(55, M, 1, B)			8
(65, F, 0, B)			8
(65,F, 1, A)			7.5
(75,M, 0, B)			9
(75,M, 1, A)			8

(Example from Uri Shalit)

(age, gender, exercise)	Y_0: Sugar levels had they received medication A	Y_1 :Sugar levels had they received medication B	Observed sugar levels
(45, F, 0)			6
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mean(sugar|medication B) – mean(sugar|medicaton A) = ?

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(75,M, 0)		9	9
(75,M, 1)	8		8

mean(sugar|medication B) – mean(sugar|medicaton A) = 7.875 - 7.125 = 0.75

(age, gender, exercise)	Y_0: Sugar levels had they received medication A	Y_1 :Sugar levels had they received medication B	Observed sugar levels
(45, F, 0)	6	5.5	6
(45, F, 1)	7	6.5	6.5
(55, M, 0)	7	6	7
(55, M, 1)	9	8	8
(65, F, 0)	8.5	8	8
(65,F, 1)	7.5	7	7.5
(75,M, 0)	10	9	9
(75,M, 1)	8	7	8

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mean(sugar|medication B) – mean(sugar|medicaton A) = 7.875 - 7.125 = 0.75

mean(sugar|had they received B) – mean(sugar|had they received A) = ?

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mean(sugar|medication B) – mean(sugar|medicaton A) = 7.875 - 7.125 = 0.75

mean(sugar|had they received B) – mean(sugar|had they received A) = 7.125 - 7.875 = -0.75

(Example from Uri Shalit)

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Typical assumption – no unmeasured confounders

- Y_0 ,Y_1 :potential outcomes for control and treated
- x: unit covariates (features)
- T: treatment assignment

- We assume:

$$(Y_0, Y_1) \perp\!\!\!\perp T \mid x$$

- The potential outcomes are independent of treatment assignment, conditioned on covariates x.

Typical assumption – no unmeasured confounders

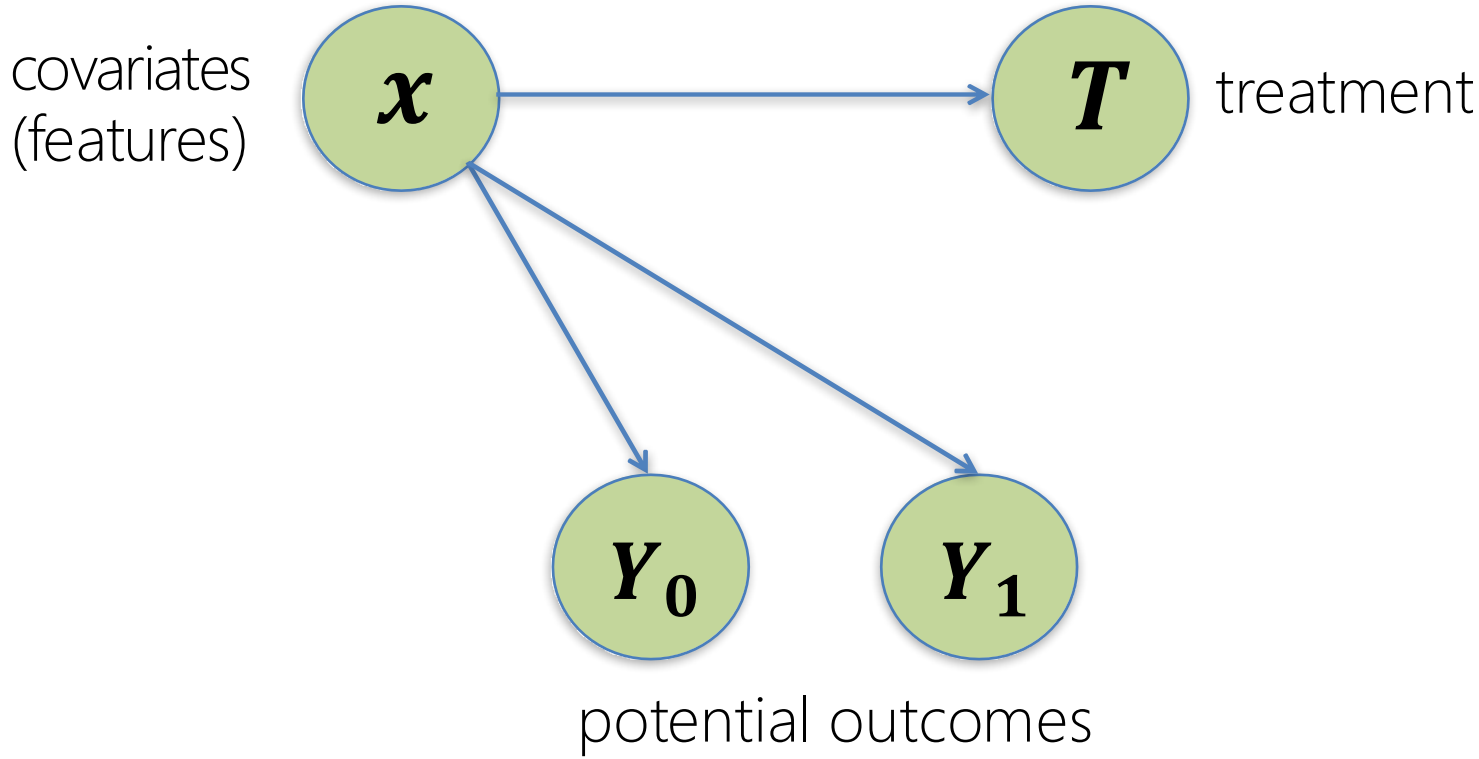
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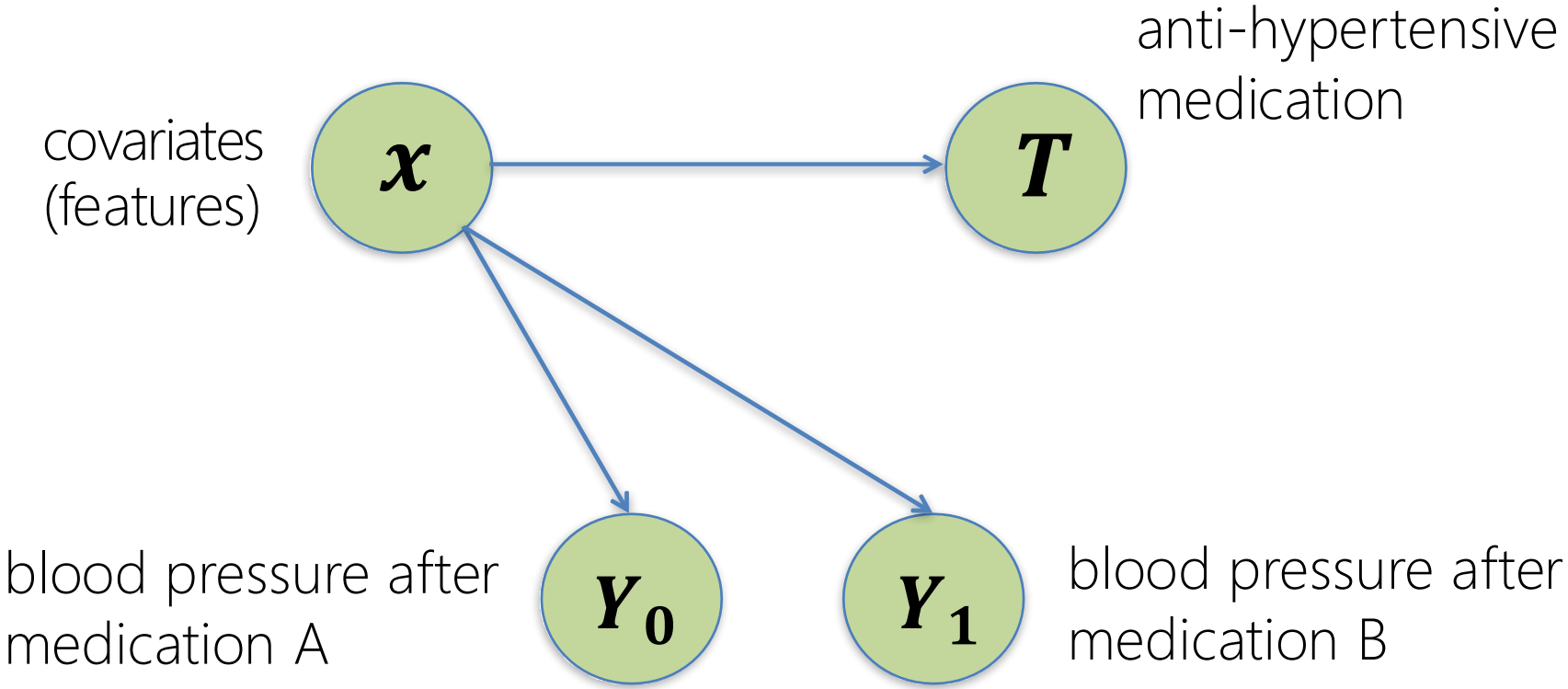
Ignorability

Ignorability



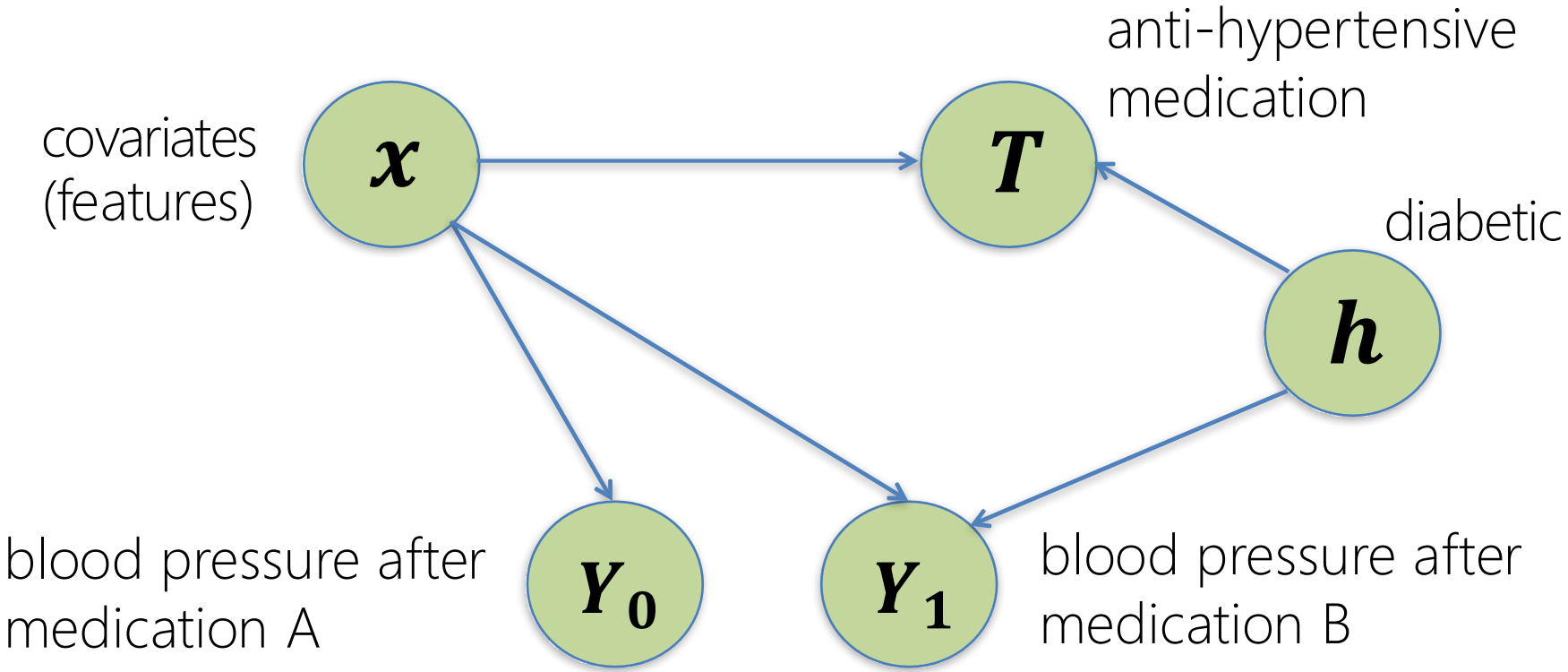
$$(Y_0, Y_1) \perp T \mid x$$

Ignorability Example



$$(Y_0, Y_1) \perp T \mid x$$

No Ignorability!!



$$(Y_0, Y_1) \not\perp T \mid x$$

Typical assumption – common support

- Y_0 ,Y_1 :potential outcomes for control and treated
- x: unit covariates (features)
- T: treatment assignment

- We assume:
  - $p(T = t|X = x) > 0 \; \forall x, t$

Causal Inference: Considerations

- What is the specific causal inference question that we are interested in?
- How to collect data to answer that question?
- What should $X, T,$ and Y be to satisfy ignorability?
- Should we worry about common support?

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Average Treatment Effect

- The expected causal effect of T on Y:

- $ATE = \mathbb{E}[Y_1 - Y_0]$

- $\mathbb{E}[Y_1] = \mathbb{E}_{x \sim p(x)}[\mathbb{E}[Y_1|x]]$ Law of total expectation
- $\mathbb{E}[Y_1] = \mathbb{E}_{x \sim p(x)}[\mathbb{E}[Y_1|x, T = 1]]$ Ignorability $((Y_0, Y_1) \perp T|x)$

Average Treatment Effect

- The expected causal effect of T on Y:

- $ATE = \mathbb{E}[Y_1 - Y_0]$

- $\mathbb{E}[Y_0] = \mathbb{E}_{x \sim p(x)}[\mathbb{E}[Y_0|x]]$ Law of total expectation
- $\mathbb{E}[Y_0] = \mathbb{E}_{x \sim p(x)}[\mathbb{E}[Y_0|x, T = 0]]$ Ignorability $((Y_0, Y_1) \perp T|x)$

Average Treatment Effect

- The expected causal effect of T on Y:

- $ATE = \mathbb{E}[Y_1 - Y_0]$

$$= \mathbb{E}_{x \sim p(x)}[\mathbb{E}[Y_1|x, T=1] - \mathbb{E}[Y_0|x, T=0]]$$

Quantities we can estimate from data

$\mathbb{E}[Y_0|x, T=1]$

$\mathbb{E}[Y_1|x, T=0]$

$\mathbb{E}[Y_0|x]$

$\mathbb{E}[Y_1|x]$

Quantities we cannot directly estimate from data

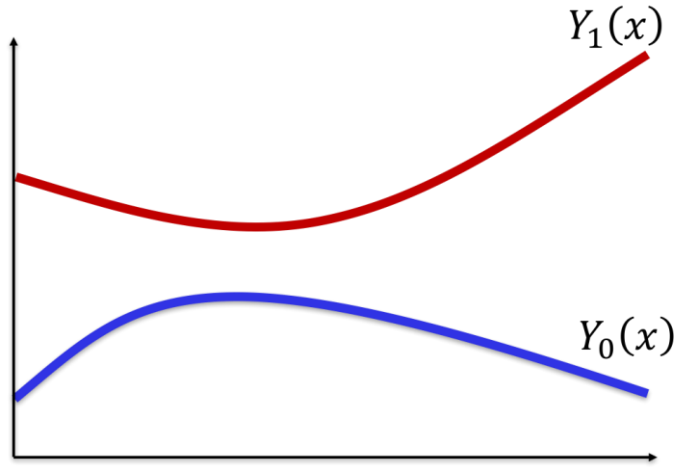
Many methods!

- Covariate adjustment
- Propensity score re-weighting
- Doubly robust estimators
- Matching
- …

Covariate adjustment

- Explicitly model the relationship between treatment, confounders, and outcome

- Also called "Response Surface Modeling"



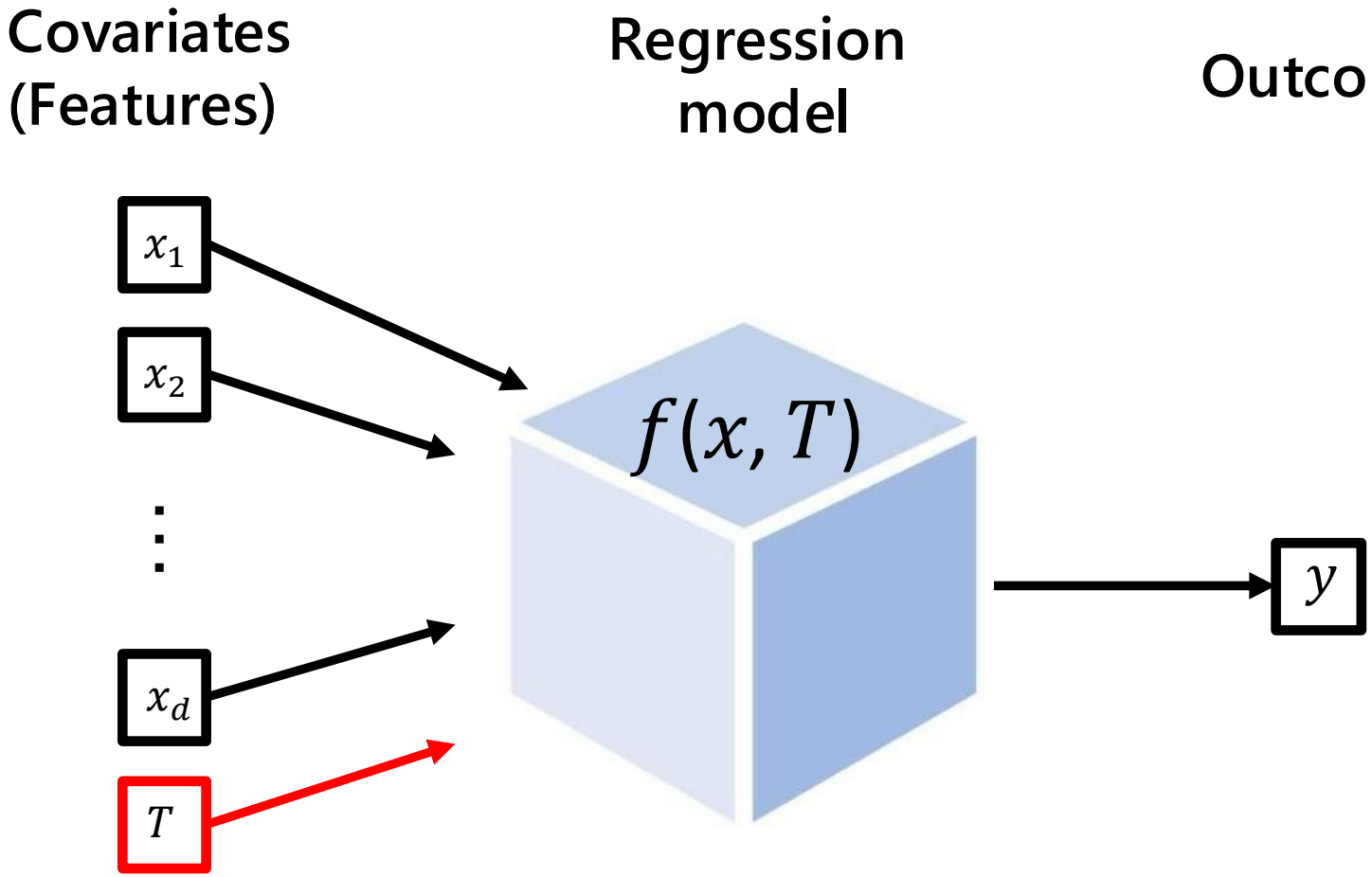
Covariate adjustment (parametric g-formula)

- Explicitly model the relationship between treatment, confounders, and outcome.
- Under ignorability, the expected causal effect of T on Y:

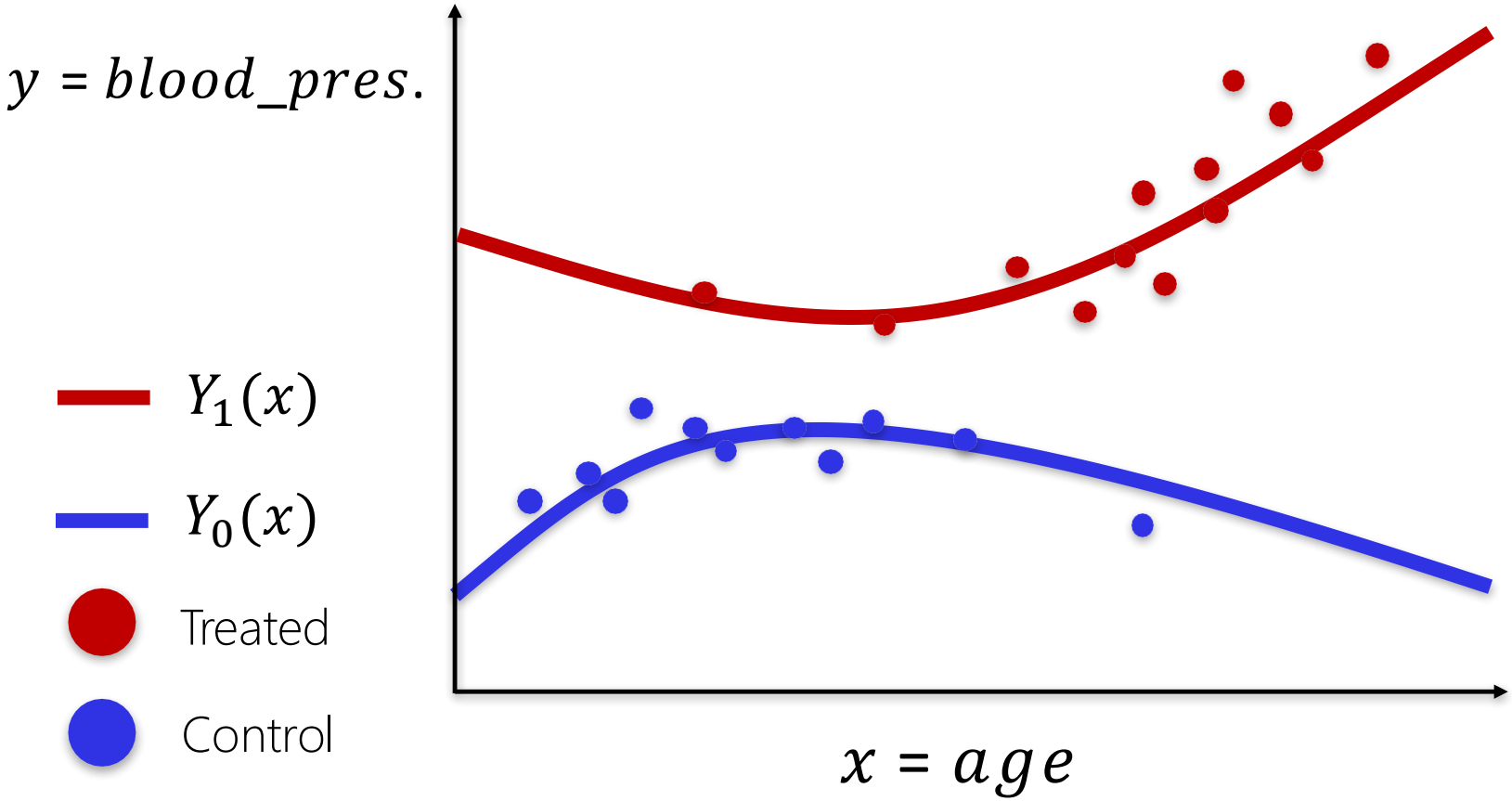
$$\mathbb{E}_{x\sim p(x)}[\mathbb{E}[Y_1|x,T=1]-\mathbb{E}[Y_0|x,T=0]]$$

- Fit a model $f(x,t)\approx\mathbb{E}[Y_t|T=t,x]$
- Then,

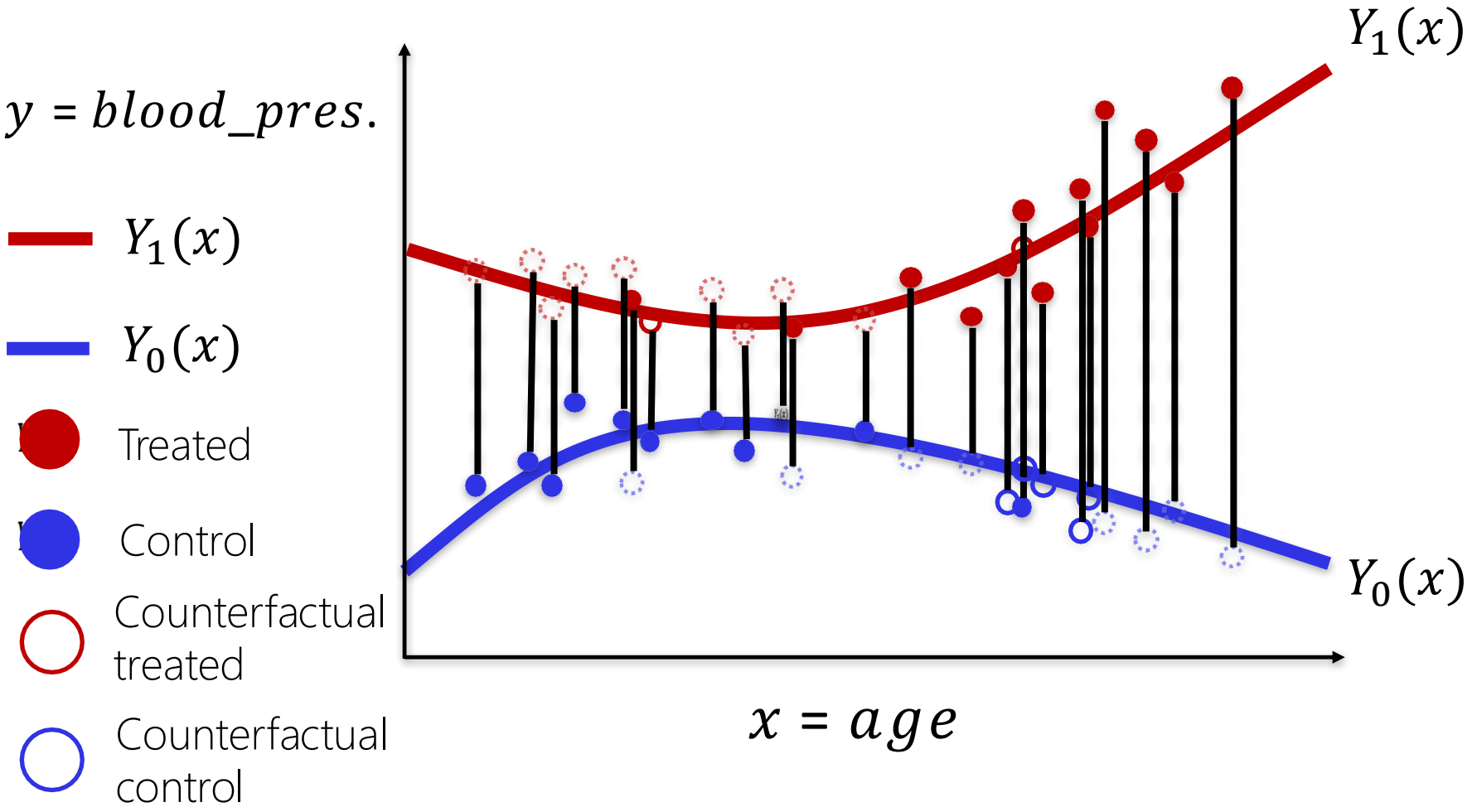
$$\widehat{ATE}=\frac{1}{n}\sum_{i=1}^{n}f(x_i,1)-f(x_i,0)$$


Covariates (Features)
Regression model
Outcome
x_1
x_2
x_d
T
$f(x, T)$
y

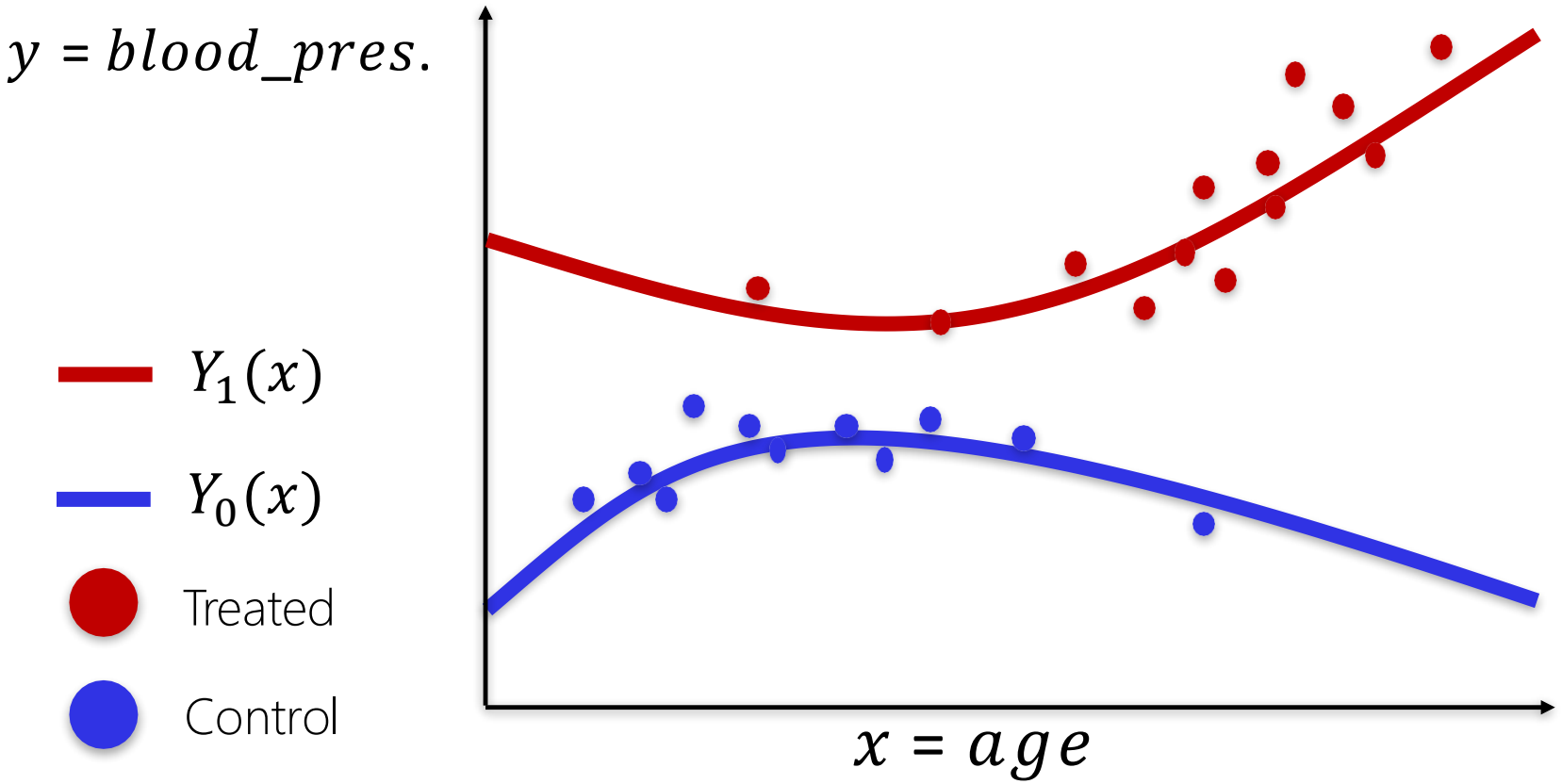
Covariate Adjustment



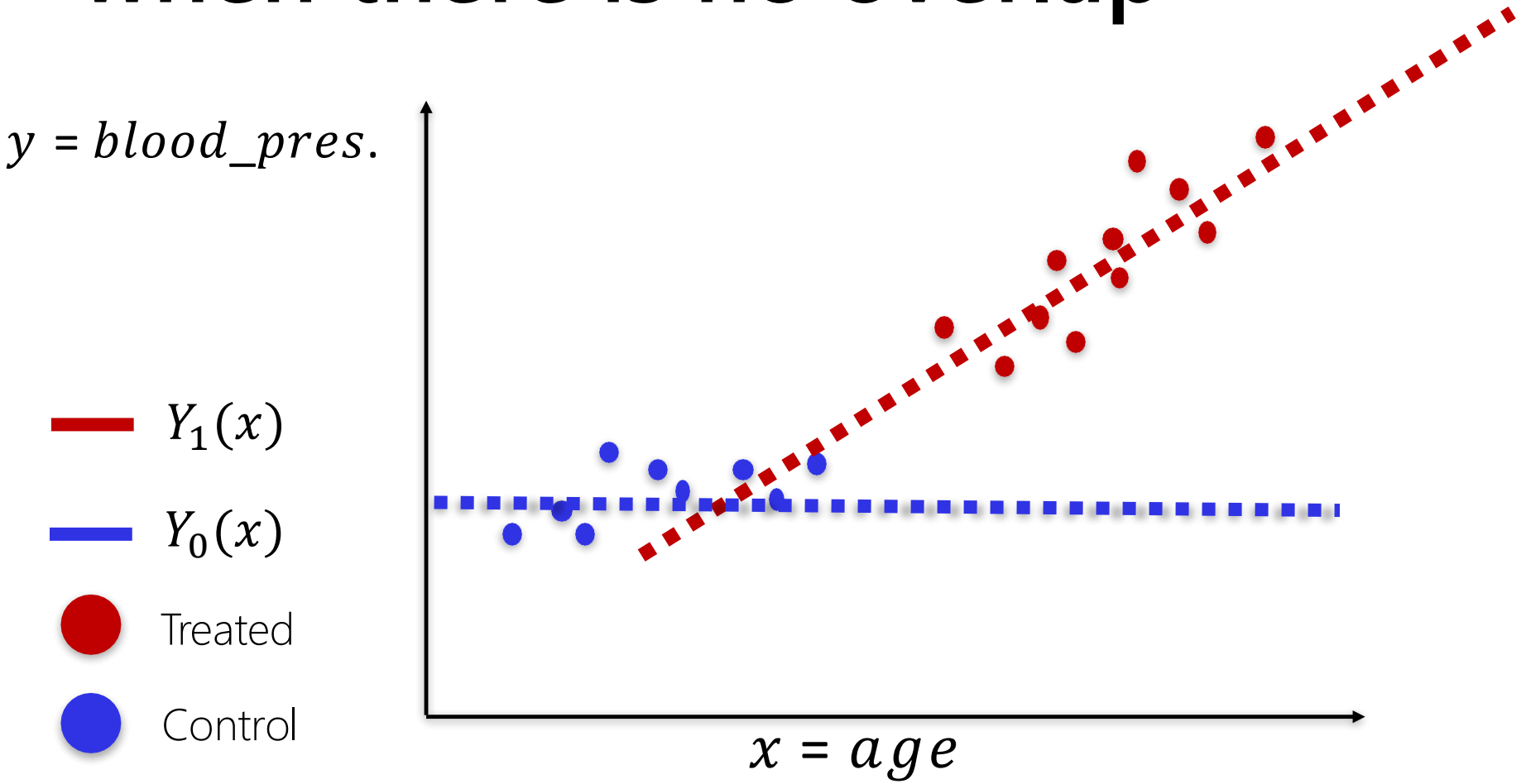
Covariate Adjustment



Covariate adjustment can fail when there is no overlap



Covariate adjustment can fail when there is no overlap



Exit pass

https://z.umn.edu/ml4h_exitpass

